

Audit of the ordinary-double-point gap route to cubic compactifications

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Abstract

This research audit records the precise deduction of the complex Gromov–Hausdorff compactification and GIT comparison for cubic n -folds from the unrestricted ordinary-double-point normalized-volume bound in every dimension $2 \leq k \leq n$. It checks the global Reeb minimum, irregular cones, the retained Cartier polarization, and the scope of the moduli objects. The deduction uses Li–Liu and Spotti–Sun; it is not a new moduli mechanism. A publicly accessible manuscript of Kong, Shen, Junyan Zhao and Minghao Zhao already states a stronger comparison in arbitrary dimension. Consequently this note is recorded in public repository checkpoints as an audit artifact, not submitted as a new solution.

Status and provenance. The upstream manuscript [1] is pinned at repository commit `adc7f1241b42` (the full hash is in the source manifest). Its theorem is a principal external input, not a theorem reproved in this note. Separate scoped adversarial audits inspected its geometric and semigroup arguments and its sole fourfold companion. These are evidence, not formal certification. No Lean verification of the gap or this consequence is claimed. AI tools were used extensively in research, drafting, and verification. This note has not undergone conventional human peer review. Public repository checkpoints preserve the owned audit findings; no Zenodo preprint submission has been made.

1 Exact target and dependency diagram

Fix $n \geq 5$ and let

$$G_n = \mathbb{P}(H^0(\mathbb{P}^{n+1}, \mathcal{O}(3))) // \mathrm{SL}(n+2).$$

The intended GH moduli object retains complex structure: its points are biholomorphic isometry classes of Kähler–Einstein cubic n -folds, with $\mathrm{Ric}(\omega) = \omega$, and its compactification is the complex/polarized GH compactification used in [3]. Bare metric-space isometry forgets complex structure and identifies conjugate varieties; it is not the moduli object in the comparison.

The good moduli space on the K side has closed complex points representing K-polystable $\mathbb{Q}$ -Fano varieties that admit a $\mathbb{Q}$ -Gorenstein deformation to a smooth cubic. Equivalently one considers the corresponding closure of the smooth cubic locus in the smoothable K-moduli space, with its complex analytic topology. Equal dimension and volume alone do not select that smoothing locus. The established smoothable KE/GH correspondence is the relevant identification with these closed points [7].

The dependency diagram for the proposed deduction is

$$\begin{array}{c}
\widehat{\text{vol}}(x, X) \leq 2(k-1)^k \quad (2 \leq k \leq n) \quad [1] \\
\Downarrow \text{ global Reeb minimum and density identity [2]} \\
\Theta(C) \leq 2(1-1/k)^k \quad \text{for isolated singular metric cones} \\
\Downarrow \text{ iteration and Cartier-root theorem [3]} \\
-K_Z = (n-1)L_Z, \quad L_Z \text{ Cartier}, \quad L_Z^n = 3 \\
\Downarrow \text{ Fujita classification; CM positivity; continuity [6, 3]} \\
\overline{\mathcal{M}}_n^{\text{GH}, \mathbb{C}} \cong_{\text{homeo}} G_n(\mathbb{C}).
\end{array}$$

Only the upper gap bound is needed; its equality classification is not used. No scheme/stack isomorphism, equivalence of moduli functors, or assertion about every nonclosed semistable orbit is deduced here.

2 The infimum-to-metric implication

Write $\widehat{\text{vol}}(x, X) = \inf_v A_X(v)^k \text{vol}(v)$ for a boundary-zero complex algebraic klt germ of dimension k . Ordinary volume is normalized by $k!$ times the leading valuation-ideal colength.

Proposition 1. *Assume the algebraic gap bound in dimension $k \geq 2$. Then an isolated singular Ricci-flat Kähler cone with compact smooth Sasaki-Einstein link has density at most $b_k = 2(1 - 1/k)^k$.*

Proof. Use the cone metric $dr^2 + r^2 g_Y$, with

$$\Theta(C(Y)) = \frac{\text{Vol}(Y)}{\text{Vol}(S^{2k-1})}.$$

The Sasakian cone has a normal affine algebraic completion; a quasi-regular CR-preserving approximation constructs it as an orbifold line-bundle cone [4, §3]. It is $\mathbb{Q}$ -Gorenstein. The positive pluricanonical Reeb weight gives klt, by [5, Lemma 6.1]. Rationality and $\mathbb{Q}$ -Gorensteinness alone are not being substituted for this klt criterion. The cone germ has zero boundary.

There is a convenient way to respect the genuine top-form hypothesis in the setup of Li-Liu §6. If $\pi_1(Y)$ is nontrivial, Myers' theorem gives a finite universal covering of degree $d \geq 2$. Bishop's inequality on that cover yields $\Theta(C(Y)) \leq 1/d \leq 1/2 \leq b_k$. If Y is simply connected, the Ricci-flat punctured cone has a global parallel holomorphic top form. Its Euler weight is k ; it extends reflexively across the vertex and is equivariant for the cone torus. This satisfies the canonical-form setup of Li-Liu.

Li-Liu's Theorem 1.9 (=6.2) states that the metric Reeb valuation v_ξ globally minimizes over *all* real valuations centered at the vertex. The theorem includes irregular Reeb fields. Their Lemma 6.8 and limiting proof use quasi-regular structures converging smoothly; these approximants are not incorrectly assumed to remain Einstein. Lemma 6.3 and the same limiting argument give

$$A_X(v_\xi) = k, \quad \widehat{\text{vol}}_X(v_\xi) = k^k \Theta(C(Y)).$$

The ratio to the round sphere cancels the fixed conversion between contact volume and Riemannian link volume. Both sides are continuous in the quasi-regular approximation. Thus, in the simply connected case,

$$k^k \Theta(C(Y)) = \widehat{\text{vol}}(o, X) \leq 2(k-1)^k.$$

An upper bound on the infimum alone would not imply this inequality for an arbitrary Reeb valuation. The global minimum is essential. $\square$

Flat $\mathbb{C}^k$ has density 1 and normalized volume k^k ; it is excluded from the singular assertion. The ODP degree valuation has discrepancy $k-1$ and volume 2, consistent with the stated normalization. No direct application of Li-Liu to a cone with singular link is made.

3 From density to polarized cubic limits

Spotti-Sun's constant $A'(n)$ is the supremum of the densities of products of a flat factor with an isolated singular cone having smooth link. Their iterated-cone argument treats other singular links [3, Lemma 5.3 and Theorem 5.2]. The required transverse dimensions are exactly $k = 2, \dots, n$. A normal klt algebraic curve is smooth, and the relevant KE singular strata have complex codimension at least two.

The sequence b_k increases. For real $t > 1$,

$$\frac{d}{dt} \log b_t = \log(1 - 1/t) + \frac{1}{t-1} = u - \log(1 + u) > 0, \quad u = \frac{1}{t-1}.$$

Products with Euclidean space do not change density, so the preceding proposition gives $A'(n) \leq b_n$.

For a smooth cubic, adjunction gives

$$-K_X = (n-1)H, \quad H^n = 3, \quad V = (-K_X)^n = 3(n-1)^n.$$

Spotti-Sun Theorem 5.2 requires

$$V > \frac{1}{2}A'(n)(n+1)^n.$$

This follows strictly from

$$\frac{1}{2}b_n(n+1)^n = (n-1)^n(1+1/n)^n < e(n-1)^n < 3(n-1)^n.$$

Its conclusion supplies both Gorenstein canonical singularities and the Cartier root L_Z with $-K_Z = (n-1)L_Z$. Preserving canonical singularities alone would not preserve the required polarization. The theorem's proof uses the polarized smoothability of the limit, the tangent-cone Cartier index comparison, and quotient-smoothing/rigidity arguments. These are inherited hypotheses and conclusions, not consequences of density alone for arbitrary $\mathbb{Q}$ -Fanos.

Fujita's Gorenstein del Pezzo classification applies to (Z, L_Z) [6]. Its intermediate-cohomology hypothesis holds for all integer twists: Kawamata-Viehweg vanishing applies to tL_Z for $t \geq 2-n$, since $tL_Z - K_Z = (t+n-1)L_Z$ is ample. Serre duality applies the same argument at twist $-t - (n-1)$ for $t \leq -1$. These ranges cover all integers for $n \geq 3$, and klt makes Z Cohen-Macaulay. The degree-three case has $h^0(Z, L_Z) = n+2$ and L_Z very ample, so it embeds Z as a cubic hypersurface in $\mathbb{P}^{n+1}$. This uses the canonical Gorenstein version, without imposing an extra terminality assumption.

4 Moduli continuity and the precise consequence

Here is an explicit positivity check for the relevant CM polarization. Put $S = \mathbb{P}(H^0(\mathbb{P}^{n+1}, \mathcal{O}(3)))$, let h and s denote the hyperplane classes on $\mathbb{P}^{n+1}$ and S , and put $r = n-1$. The universal cubic has class $3h + s$, and its relative anticanonical class is $rh - s$. In the standard Fano normalization,

$$\begin{aligned} c_1(\lambda_{\text{CM}}) &= -\pi_*((rh - s)^{n+1}(3h + s)) \\ &= r^n(3(n+1) - r)s = 2(n+2)(n-1)^n s. \end{aligned}$$

The expression is computed on the product; the universal-cubic factor implements pushforward from the hypersurface. Any positive conventional rescaling preserves the sign. The coefficient is strictly positive, so the CM polarization agrees in sign with the standard GIT polarization.

Spotti–Sun §5.2 applies their moduli continuity method from §4.2. The initial Fermat cubic is KE; the smooth cubic moduli locus is connected. KE existence is open there, and the cubic-limit and CM facts make it closed. The map from the complex GH compactification to GIT is continuous; uniqueness of the KE metric on a fixed Fano variety gives injectivity. Its compact image contains the dense smooth GIT locus, giving surjectivity. Compactness and the Hausdorff analytic GIT topology give a homeomorphism. This is precisely the established transfer [3, Theorem 1.3(2)], rather than a new continuity theorem.

Corollary 2 (Audited conditional consequence). *If the unrestricted algebraic normalized-volume gap holds in every dimension $2 \leq k \leq n$, then for $n \geq 5$ the complex GH compactification of smooth KE cubic n -folds is naturally homeomorphic to $G_n(\mathbb{C})$. Its closed points also identify with the corresponding smoothable K -polystable Fano limits in the cubic smoothing locus.*

Taking the upper-bound theorem of [1] as the external input removes the dimension-by-dimension metric-gap hypothesis in [3, Theorem 1.3(2)]. That is the entire proposed moduli contribution. The topological result alone does not upgrade this deduction to schemes, stacks, or nonclosed semistable points.

5 Priority finding and publication decision

The lower-dimensional comparison is prior work: cubic surfaces by Odaka–Spotti–Sun, cubic threefolds by Liu–Xu, and cubic fourfolds by Liu [8, 9, 10]. Cubic curves are not included in a positive-KE Fano headline. Smooth KE existence should not be advertised as newly discovered here.

The primary author page [11] currently links a September 30, 2026 manuscript, publicly downloaded on October 6, 2026 (Pacific time). Its Theorem 1.1 states the arbitrary-dimensional cubic comparison, including stacks and good moduli spaces; its proof uses refined local inequalities and tangent-sheaf semistability. The actual earliest web posting is unverified, and a manuscript date alone does not establish public priority. Its present public availability is verified. This audit does not independently certify every lemma of that separate paper.

The requested theorem is therefore already publicly claimed in stronger form. Moreover, the proposed alternative mechanism is the known Spotti–Sun reduction with a new upstream input, whose gap proof belongs to OpenAI. No genuinely new in-scope extension has been established in this effort. The evidence supports a useful checked deduction and source audit; it does not support advertising a new resolution. In accordance with the project’s priority conditions, no Zenodo deposit or DOI-tracker entry is made. The persistent publication objective remains unachieved.

References

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